python code:

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# Student Record Management System using Singly Linked List

# Node class
class Node:
    def __init__(self, roll_no, name, marks):
        self.roll_no = roll_no
        self.name = name
        self.marks = marks
        self.next = None

# Linked List class
class StudentLinkedList:
    def __init__(self):
        self.head = None

    # Add student
    def add(self, roll_no, name, marks):
        new_node = Node(roll_no, name, marks)
        if self.head is None:
            self.head = new_node
        else:
            temp = self.head
            while temp.next is not None:
                temp = temp.next
            temp.next = new_node
        print("Student record added successfully.")

    # Display students
    def display(self):
        if self.head is None:
            print("No student records available.")
            return
        print("\n----- STUDENT RECORDS -----")
        temp = self.head
        while temp is not None:
            print("Roll No:", temp.roll_no)
            print("Name:", temp.name)
            print("Marks:", temp.marks)
            print("----------------------")
            temp = temp.next
```

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# Search student
def search(self, roll_no):
    temp = self.head
    while temp is not None:
        if temp.roll_no == roll_no:
            print("\nStudent Found")
            print("Roll No:", temp.roll_no)
            print("Name:", temp.name)
            print("Marks:", temp.marks)
            return
        temp = temp.next
    print("Student record not found.")

# Sort records
def sort(self, choice, order):
    if self.head is None:
        print("List is empty.")
        return

    current = self.head
    while current is not None:
        next_node = current.next
        while next_node is not None:
            if choice == 1:
                value1 = current.roll_no
                value2 = next_node.roll_no
            else:
                value1 = current.marks
                value2 = next_node.marks

            # Ascending
            if order == 1 and value1 > value2:
                self.swap_data(current, next_node)

            # Descending
            elif order == 2 and value1 < value2:
                self.swap_data(current, next_node)

            next_node = next_node.next
        current = current.next
    print("Records sorted successfully")
```

```
# Swap student data
def swap_data(self, node1, node2):
    node1.roll_no, node2.roll_no = node2.roll_no, node1.roll_no
    node1.name, node2.name = node2.name, node1.name
    node1.marks, node2.marks = node2.marks, node1.marks

# Delete student
def delete(self, roll_no):
    if self.head is None:
        print("List is empty.")
        return

    # Delete first node
    if self.head.roll_no == roll_no:
        self.head = self.head.next
        print("Student record deleted successfully.")
        return

    temp = self.head
    while temp.next is not None:
        if temp.next.roll_no == roll_no:
            temp.next = temp.next.next
            print("Student record deleted successfully.")
            return
        temp = temp.next
    print("Student record not found.")

# Update student
def update(self, roll_no):
    temp = self.head
    while temp is not None:
        if temp.roll_no == roll_no:
            print("Current Name:", temp.name)
            print("Current Marks:", temp.marks)
            temp.name = input("Enter new name: ")
            temp.marks = float(input("Enter new marks: "))
            print("Student record updated successfully.")
            return
        temp = temp.next
    print("Student record not found.")
```

```
# Main program
student_list = StudentLinkedList()

while True:
    print("\n====== STUDENT RECORD MANAGEMENT ======")
    print("1. Add Student")
    print("2. Delete Student")
    print("3. Update Student")
    print("4. Search Student")
    print("5. Display Students")
    print("6. Sort Records")
    print("7. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        roll_no = int(input("Enter Roll No: "))
        name = input("Enter Name: ")
        marks = float(input("Enter Marks: "))
        student_list.add(roll_no, name, marks)

    elif choice == 2:
        roll_no = int(input("Enter Roll No to delete: "))
        student_list.delete(roll_no)

    elif choice == 3:
        roll_no = int(input("Enter Roll No to update: "))
        student_list.update(roll_no)

    elif choice == 4:
        roll_no = int(input("Enter Roll No to search: "))
        student_list.search(roll_no)

    elif choice == 5:
        student_list.display()

    elif choice == 6:
        print("\nSort By:")
        print("1. Roll Number")
        print("2. Marks")
        sort_choice = int(input("Enter choice: "))
        print("\nOrder:")
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        print("1. Ascending")
        print("2. Descending")
        order = int(input("Enter order: "))
        student_list.sort(sort_choice, order)
        student_list.display()

    elif choice == 7:
        print("Program terminated.")
        break

    else:
        print("Invalid choice!")
```

OUTPUT:

```
====== STUDENT RECORD MANAGEMENT ======
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Display Students
6. Sort Records
7. Exit
Enter your choice: 1
Enter Roll No: 101
Enter Name: Raj
Enter Marks: 85.5
Student record added successfully.

====== STUDENT RECORD MANAGEMENT ======
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Display Students
6. Sort Records
7. Exit
Enter your choice: 1
Enter Roll No: 102
Enter Name: Priya
```

Enter Marks: 92.0
Student record added successfully.

====== STUDENT RECORD MANAGEMENT ======
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Display Students
6. Sort Records
7. Exit
Enter your choice: 1
Enter Roll No: 103
Enter Name: Aditya
Enter Marks: 78.5
Student record added successfully.

====== STUDENT RECORD MANAGEMENT ======
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Display Students
6. Sort Records
7. Exit
Enter your choice: 5

----- STUDENT RECORDS -----
Roll No: 101
Name: Raj
Marks: 85.5

Roll No: 102
Name: Priya
Marks: 92.0

Roll No: 103
Name: Aditya
Marks: 78.5

====== STUDENT RECORD MANAGEMENT ======

1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Display Students
6. Sort Records
7. Exit
Enter your choice: 4
Enter Roll No to search: 102

Student Found
Roll No: 102
Name: Priya
Marks: 92.0

====== STUDENT RECORD MANAGEMENT ======
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Display Students
6. Sort Records
7. Exit
Enter your choice: 6

Sort By:
1. Roll Number
2. Marks
Enter choice: 2

Order:
1. Ascending
2. Descending
Enter order: 1
Records sorted successfully

----- STUDENT RECORDS -----
Roll No: 103
Name: Aditya
Marks: 78.5

Roll No: 101

Name: Raj
Marks: 85.5

Roll No: 102
Name: Priya
Marks: 92.0

====== STUDENT RECORD MANAGEMENT ======
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Display Students
6. Sort Records
7. Exit
Enter your choice: 3
Enter Roll No to update: 101
Current Name: Raj
Current Marks: 85.5
Enter new name: Rajesh
Enter new marks: 88.0
Student record updated successfully.

====== STUDENT RECORD MANAGEMENT ======
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Display Students
6. Sort Records
7. Exit
Enter your choice: 2
Enter Roll No to delete: 103
Student record deleted successfully.

====== STUDENT RECORD MANAGEMENT ======
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Display Students

6. Sort Records
7. Exit
Enter your choice: 5

----- STUDENT RECORDS -----
Roll No: 101
Name: Rajesh
Marks: 88.0

Roll No: 102
Name: Priya
Marks: 92.0

====== STUDENT RECORD MANAGEMENT ======
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Display Students
6. Sort Records
7. Exit
Enter your choice: 7
Program terminated.